

🚀 Mastering Auto Scaling with Load Balancer on AWS 🌐

💡 Effortlessly Scale Your EC2 Instances for High Availability and Resilience ↻



Setting up an **Auto Scaling Group** with an **Application Load Balancer (ALB)** in AWS ensures that your application can handle varying traffic loads by automatically adjusting the number of EC2 instances. Here's a step-by-step guide to achieve this configuration:

Prerequisites

Before you begin, ensure you have:

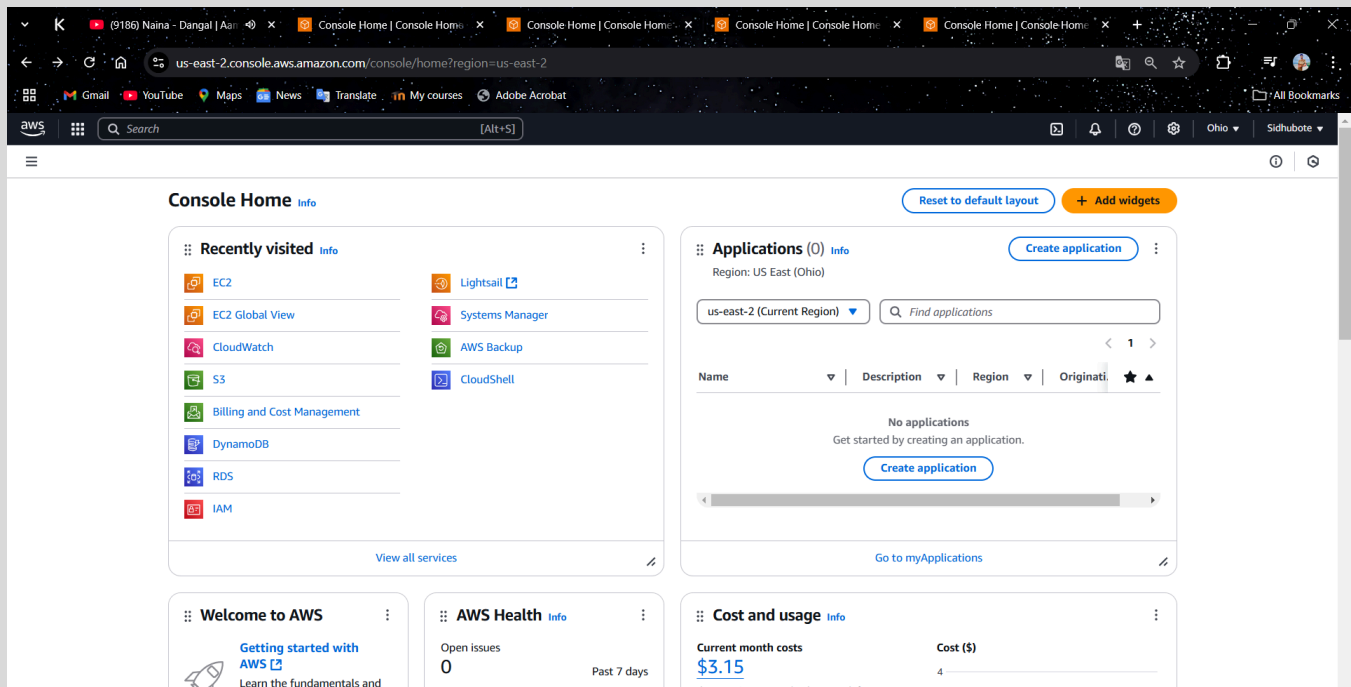
- **AWS Account:** An active AWS account with necessary permissions.
 - **Virtual Private Cloud (VPC):** Utilize the default VPC or a custom VPC with subnets in multiple Availability Zones.
 - **Security Groups:**
 - **Load Balancer Security Group:** Allows inbound HTTP (port 80) traffic.
 - **Instance Security Group:** Allows inbound traffic from the Load Balancer on the application port (e.g., port 80) and SSH (port 22) for management.
 - **IAM Role (Optional):** An IAM role that grants your application access to AWS resources.
 - **Amazon Machine Image (AMI):** A custom or default AMI to launch EC2 instances.
-

Step 1: Create a Launch Template

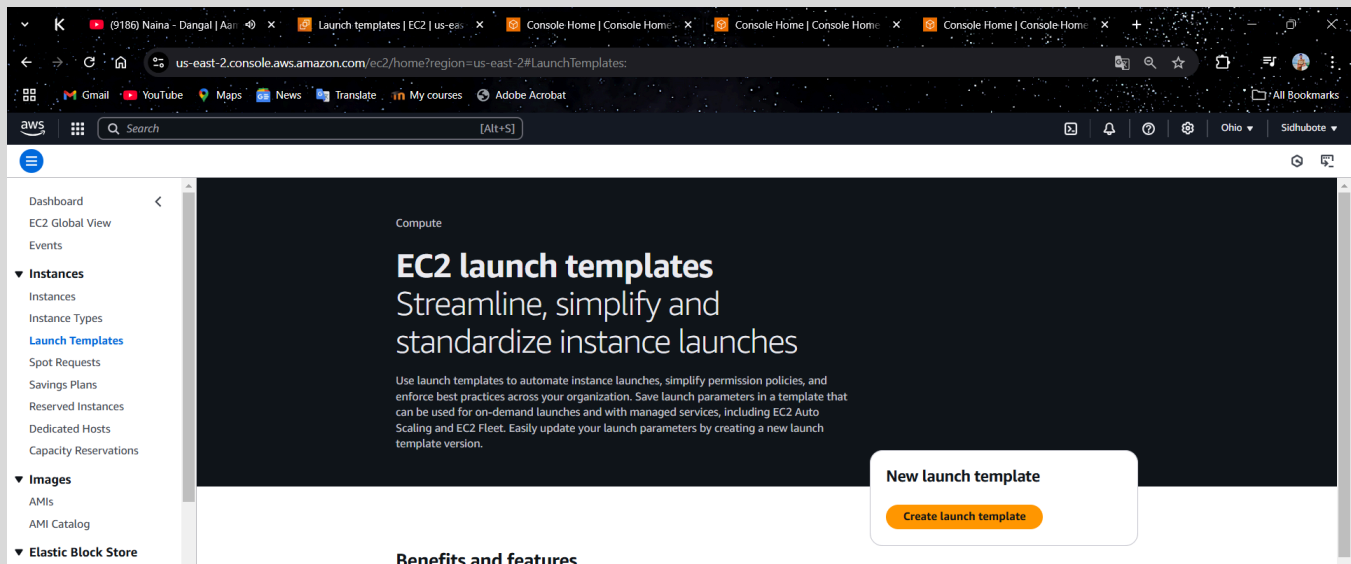
A Launch Template specifies the configuration for EC2 instances within the Auto Scaling Group.

1. Access the EC2 Dashboard:

- Sign in to the AWS Management Console.



- Navigate to **EC2 Dashboard > Launch Templates**.



2. Create Launch Template:

- Click **Create launch template**.

- Provide a **Launch template name** and **Description**.

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► **Template tags**

► **Source template**

- **Source AMI:** Select your desired AMI.

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

🔍 Search our full catalog including 1000s of application and OS images

Recents Quick Start

Don't include in launch template

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-036841078a4b68e14 (64-bit (x86)) / ami-0560690593473ded1 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

64-bit (x86) ▼

AMI ID

ami-036841078a4b68e14

Username ⓘ

ubuntu

Verified provider

- **Instance type:** Choose an appropriate instance type (e.g., **t2.micro**).

▼ Instance type [Info](#) | [Get advice](#)

Advanced

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

▼

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- **Key pair:** Select an existing key pair or create a new one for SSH access.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

kucl24

▼

↻

[Create new key pair](#)

- **Network settings:**
 - **Security groups:** Select the Instance Security Group.

▼ Network settings [Info](#)

Subnet | [Info](#)

Don't include in launch template

▼

↻ [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups | [Info](#)

Select security groups

▼

default sg-0141632d2dd9484fe ✕

VPC: vpc-0bcdd90e4cd983c9

↻ [Compare security group rules](#)

▶ Advanced network configuration

- **Storage (volumes):** Configure storage as needed.
- **Advanced details:**

▼ **Advanced details** [Info](#)

IAM instance profile [Info](#)

Don't include in launch template ▼

↻ [Create new IAM profile](#)

Hostname type [Info](#)

Don't include in launch template ▼

DNS Hostname [Info](#)

- **IAM instance profile:** Select an IAM role if applicable.
- **User data:** Add initialization scripts (e.g., to install web servers or applications).

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

⬆ Choose file

```
#!/bin/bash
sudo apt update -y
sudo apt install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
sudo apt install unzip -y
sudo wget https://www.free-css.com/assets/files/free-css-templates/download/page296/carvilla.zip
ls
sudo unzip carvilla.zip
sudo mv carvilla-v1.0/* /var/www/html
sudo systemctl restart apache2
```

☐ User data has already been base64 encoded

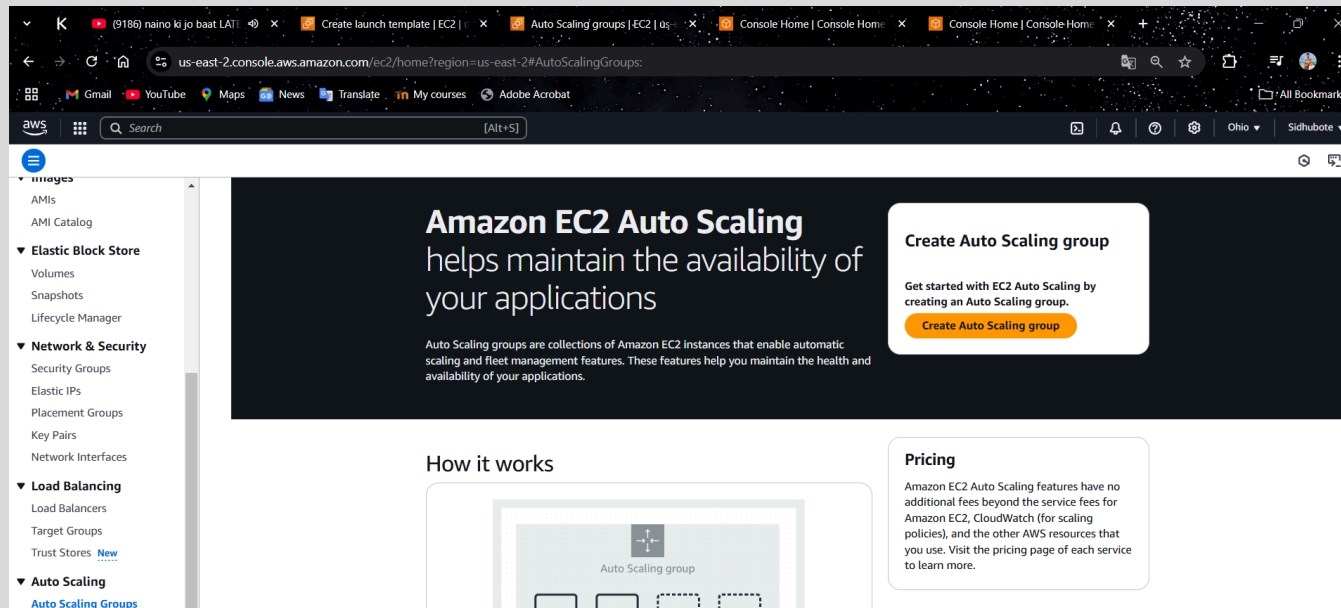
- Click **Create launch template**.

Step 2: Create an Auto Scaling Group

An Auto Scaling Group ensures the right number of EC2 instances are running to handle the load for your application.

1. Access Auto Scaling Groups:

- In the EC2 Dashboard, navigate to **Auto Scaling Groups**.



2. Create Auto Scaling Group:

- Click **Create Auto Scaling group**.

- **Name:** Enter a name for the group.

Step 1
 ● **Choose launch template** Info

Step 2
 ○ Choose instance launch options

Step 3 - optional
 ○ Integrate with other services

Step 4 - optional
 ○ Configure group size and scaling

Step 5 - optional
 ○ Add notifications

Choose launch template

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
 Enter a name to identify the group.

Autoscalegrp

Must be unique to this account in the current Region and no more than 255 characters.

- **Launch template:**
 - Select the launch template created earlier.
- Click **Next**.

○ Add notifications

Step 6 - optional
 ○ Add tags

Step 7
 ○ Review

Launch template

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
 Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

mytemp

Create a launch template

Version
 Default (1)

Create a launch template version

Description 1	Launch template mytemp lt-0240124bea9740e81	Instance type t2.micro
AMI ID ami-036841078a4b68e14	Security groups -	Request Spot Instances No
Key pair name kuc124	Security group IDs sg-0141632d2dd9484fe	

Additional details

Storage (volumes) -	Date created Sun Dec 29 2024 19:10:17 GMT+0530 (India Standard Time)
-------------------------------	--

Cancel Next

3. Configure Network:

- **VPC:** Select your VPC.

- **Subnets:** Choose the same subnets used for the ALB.
- Click **Next**.

Step 7
Review

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-0bcedd90e4cd983c9
172.31.0.0/16 Default

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-2a | subnet-0d25c37210724fcd
172.31.0.0/20 Default

us-east-2b | subnet-0ccc04079ba5fa530
172.31.16.0/20 Default

us-east-2c | subnet-0a0e598e8af231b73
172.31.32.0/20 Default

[Create a subnet](#)

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel Skip to review Previous **Next**

4. Attach Load Balancer:

- **Load balancing:**
 - Select **Attach to a new load balancer**.

Step 1
Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Integrate with other services - optional Info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ **Attach to an existing load balancer**
Choose from your existing load balancers.

☒ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

- **Load balancer type:** Choose **Application Load Balancer**.
- **Name:** Enter the ALB name.

Attach to a new load balancer

Define a new load balancer to create for attachment to this Auto Scaling group.

Load balancer type

Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, visit the [Load Balancing console](#).

☒ Application Load Balancer
HTTP, HTTPS

☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name

Name cannot be changed after the load balancer is created.

mylb-1

Load balancer scheme

Scheme cannot be changed after the load balancer is created.

☐ Internal

☒ Internet-facing

Network mapping

Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.

VPC

vpc-0bcdd90e4cd983c9

Availability Zones and subnets

You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

☒ us-east-2a

subnet-0d25c37210724fced

☒ us-east-2b

subnet-0ccc04079ba5fa530

☒ us-east-2c

subnet-0a0e598e8af231b73

- **Listeners:** Ensure HTTP (80) is selected.
- **Target group:**
 - Select the target group created earlier.

Listeners and routing

If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol

HTTP

Port

80

Default routing (forward to)

Create a target group

New target group name

An instance target group with default settings will be created.

mylb-1

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

- ☒ **No VPC Lattice service**
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

- ☐ **Attach to VPC Lattice service**
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#) [↗](#)

Application Recovery Controller (ARC) zonal shift - *new* [Info](#)

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

- ☐ **Enable zonal shift**
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - *optional* [Info](#)

- ☒ **Turn on Elastic Load Balancing health checks** Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

[i](#) EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the [Load Balancer console](#) [↗](#)

- ☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.
- ☐ **Turn on Amazon EBS health checks**
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

seconds

[Cancel](#)

[Skip to review](#)

[Previous](#)

[Next](#)

- Click **Next**.

5. Configure Scaling Policies:

- **Desired capacity:** Set the initial number of instances (e.g., 2).

The screenshot shows the 'Configure group size and scaling' step in the AWS IAM console. On the left, a vertical list of steps is shown, with 'Configure group size and scaling' highlighted. The main content area is titled 'Configure group size and scaling - optional' and includes an 'Info' icon. Below the title, a description states: 'Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.' The 'Group size' section explains that it sets the initial size of the Auto Scaling group and that it can be changed later. The 'Desired capacity type' section allows choosing the unit of measurement (vCPUs or Memory) for the desired capacity value. A dropdown menu is set to 'Units (number of instances)'. The 'Desired capacity' section has a text input field with the value '2' and a spinner icon.

- **Minimum capacity:** Set the minimum number of instances (e.g., 1).
- **Maximum capacity:** Set the maximum number of instances (e.g., 4).

The screenshot shows the 'Scaling' section in the AWS IAM console. It includes an 'Info' icon and a description: 'You can resize your Auto Scaling group manually or automatically to meet changes in demand.' The 'Scaling limits' section explains that it sets limits on how much the desired capacity can be increased or decreased. There are two input fields: 'Min desired capacity' with the value '1' and 'Max desired capacity' with the value '5'. Below each input field, a note specifies the relationship to the desired capacity: 'Equal or less than desired capacity' for the minimum and 'Equal or greater than desired capacity' for the maximum.

- **Scaling policies:**
 - Choose **Target tracking scaling policy**.
 - **Metric type:** Select **Average CPU utilization**.

- **Target value:** Set a target (e.g., 40%).

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization ▼

Target value

40

Instance warmup [Info](#)

120

seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch

Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

- Click **Next**.

7. Additional settings

● Additional capacity setting may default

Additional capacity settings

Capacity Reservation preference | [Info](#)

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

- ☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.
- ☐ **None**
Instances will not be launched into a Capacity Reservation.
- ☐ **Capacity Reservations only**
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
- ☐ **Capacity Reservations first**
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

- ☐ Enable instance scale-in protection

Monitoring | [Info](#)

- ☒ Enable group metrics collection within CloudWatch

Default instance warmup | [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

- ☐ Enable default instance warmup

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

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Monitoring | [Info](#)

- ☒ Enable group metrics collection within CloudWatch

Default instance warmup | [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

- ☐ Enable default instance warmup

[Cancel](#)[Skip to review](#)[Previous](#)[Next](#)

Add notifications - optional [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

[Add notification](#)[Cancel](#)[Skip to review](#)[Previous](#)[Next](#)

Add tags - optional [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

i You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

X

Tags (0)

Add tag

50 remaining

Cancel Previous Next

● Review all configuration and next.

Review [Info](#)

Step 1: Choose launch template [Edit](#)

Group details

Auto Scaling group name

Autoscalegrp

Launch template

Launch template	Version	Description
mytemp ↗	Default	1

Step 2: Choose instance launch options [Edit](#)

Network

VPC

vpc-0bcdd90e4cd983c9 [↗](#)

Availability Zones and subnets

Availability Zone	Subnet	Subnet CIDR range
us-east-2a	subnet-0d25c37210724fced ↗	172.31.0.0/20
us-east-2b	subnet-0ccc04079ba5fa530 ↗	172.31.16.0/20
us-east-2c	subnet-0a0e598e8af231b73 ↗	172.31.32.0/20

Availability Zone distribution

Balanced best effort

Instance type requirements

This Auto Scaling group will adhere to the launch template.

Step 3: Integrate with other services [Edit](#)

Load balancing

Load balancer 1

Name	Type	Target group
mylb-1	Application/HTTP	mytargetgrp-1

VPC Lattice integration options

VPC Lattice target groups

-

Application Recovery Controller (ARC) zonal shift

ARC zonal shift

Disabled

Health checks

Health check type

EC2, ELB

Health check grace period

300 seconds

Step 4: Configure group size and scaling policies [Edit](#)

Group size

Desired capacity

2

Desired capacity type

Units (number of instances)

Scaling

Minimum desired capacity

1

Maximum desired capacity

5

Target tracking policy

Policy type

Target tracking scaling

Scaling policy name

Target Tracking Policy

Execute policy when

AS required to maintain Average CPU utilization at 40

Take the action

Add or remove capacity units as required

Instances need

120 seconds to warm up before including in metric

Scale in

Enabled

Instance maintenance policy

Replacement behavior

No policy

Min healthy percentage

-

Max healthy percentage

-

Additional settings

Instance scale-in protection

Disabled

Monitoring

Enabled

Default instance warmup

Disabled

Capacity Reservation preference

Preference

Default

Capacity Reservation IDs

-

Resource Groups

-

Step 5: Add notifications [Edit](#)

Notifications

No notifications

Step 6: Add tags [Edit](#)

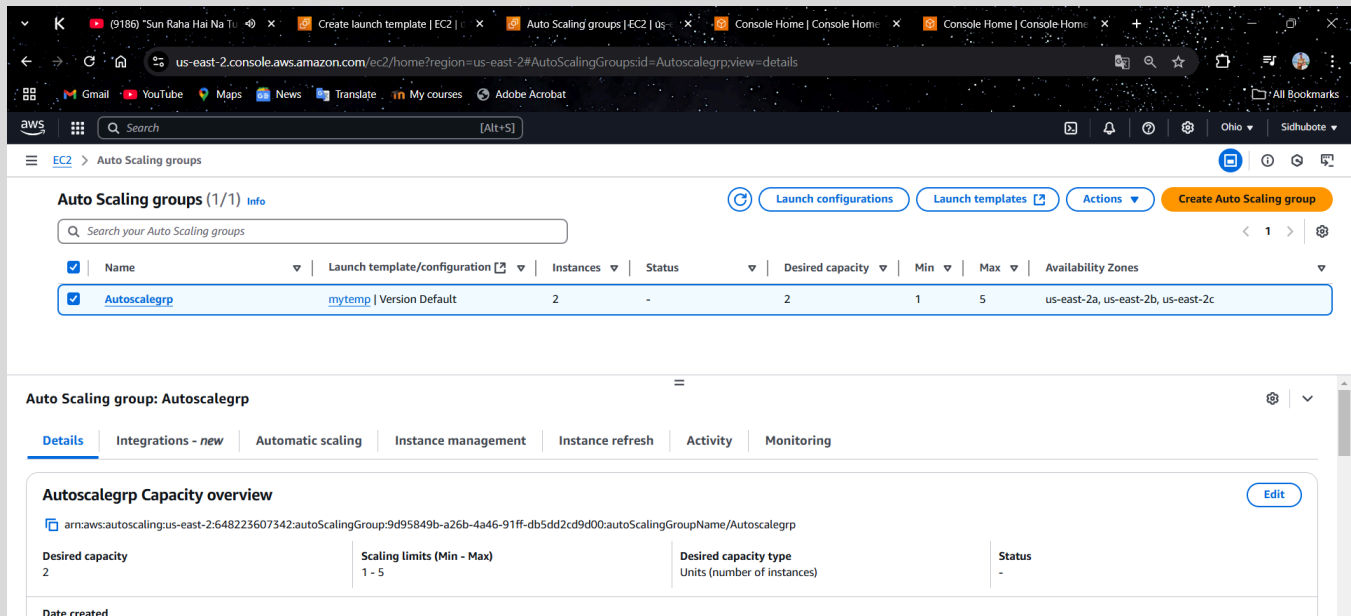
Tags (0)

Key	Value	Tag new instances
No tags		

[Preview code](#)

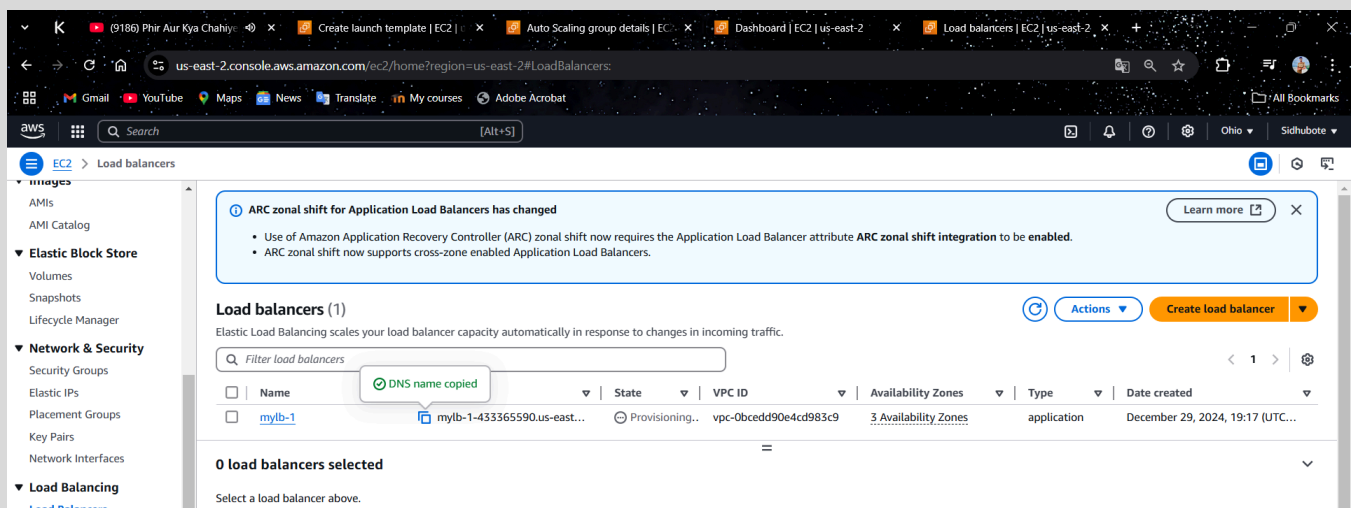
Cancel Previous Create Auto Scaling group

• Click on create auto scaling group

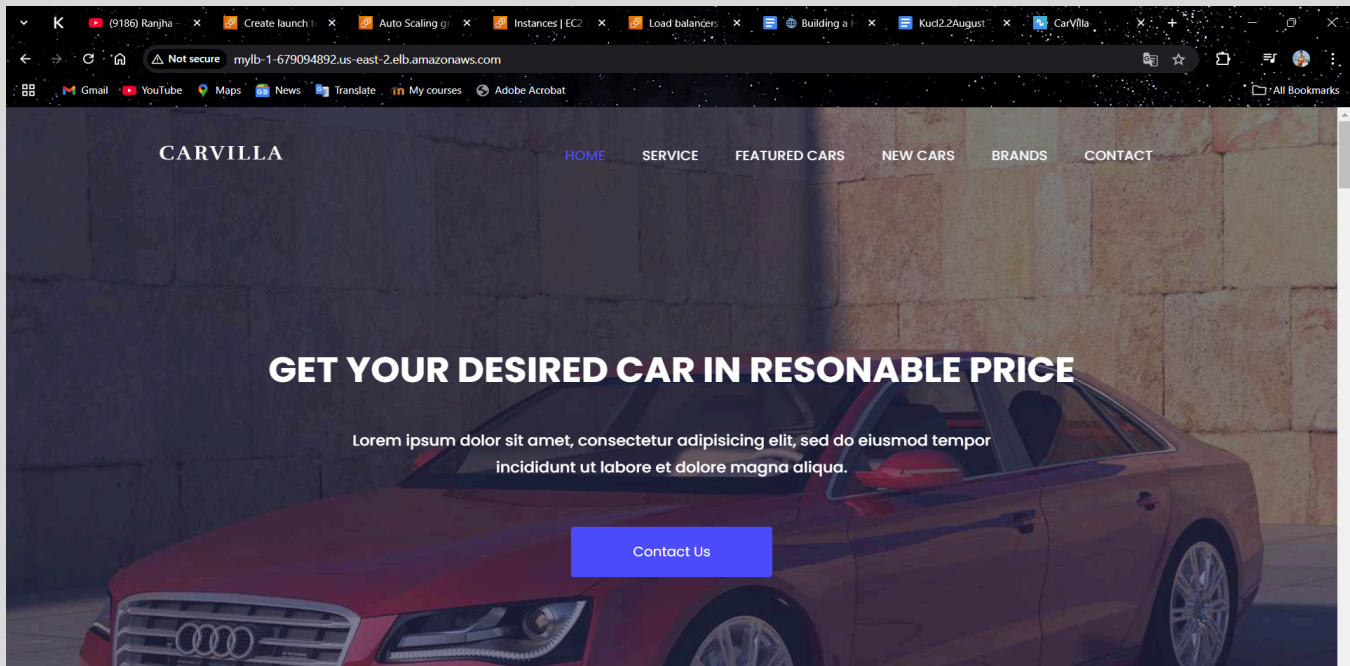


• Verify the configuration

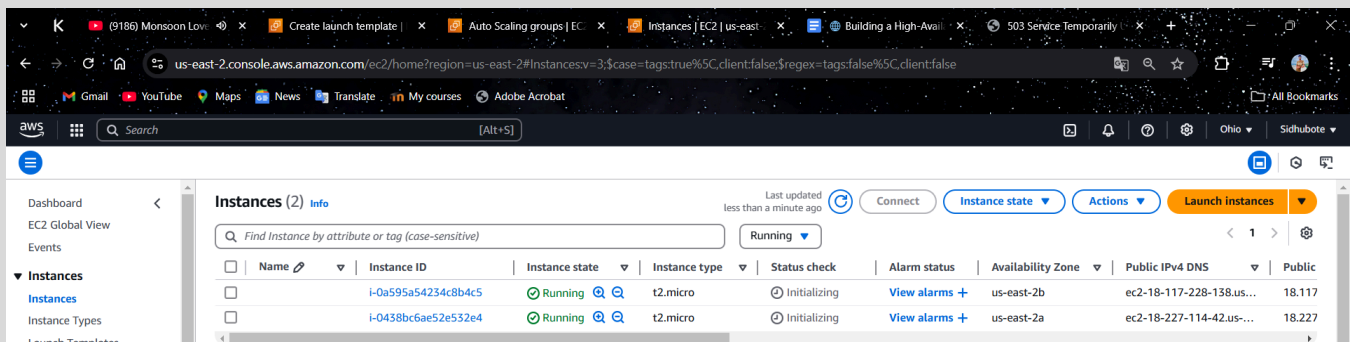
- Copy the DNS of load balancer and browse.



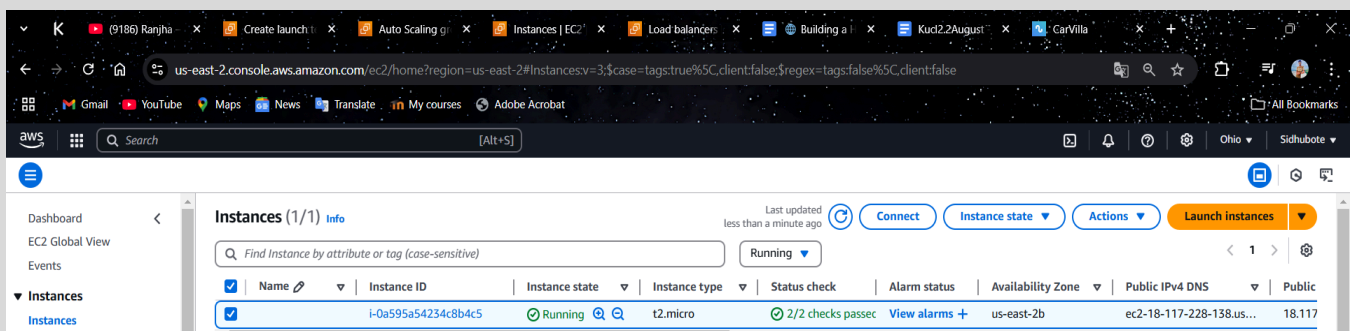
• Now ,you can see hosted website



• Check launched instances



• If there is no traffic then it will auto scale-in



us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#AutoScalingGroups:

Auto Scaling groups (1) Info

Search your Auto Scaling groups

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
☐	Autoscalegrp	mytemp Version Default	1	-	1	1	5	us-east-2a, us-east-2b, us-east-2c

